

ENVS1017/ENVS6202 Scientific Report

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Introduction

The urban coastal catchments hold streams, wetlands, estuaries and lagoons which receive the runoff of the concentrated residential, industrial and transport land uses. In these systems, urbanisation replaces permeable soil and vegetation with roofs, roads and drains, increasing impervious cover and rapidly routing stormwater to channels (Paul & Meyer, 2001; Chin et al., 2013). This modified city hydrology will generally result in flashier flows, shorter lag times, larger peak discharges, lower infiltration and more frequent bankfull or erosive events, which promote channel enlargement and instability (Booth, 2000; Walsh et al., 2012). Poor water quality, such as high turbidity and suspended solids, nutrients, metals and other road, lawn and golf course and industrial wastes, is also a common feature of urban streams (Preston, 1995; Birch et al., 2021). The combined hydrologic, geomorphic and water-quality degradation is widely described as the urban stream syndrome (Walsh et al., 2005). Aquatic health here refers to the capacity of a waterway to maintain ecological structure, water quality and habitat which support aquatic biota.

These problems are significant to Manly Lagoon since Brookvale Creek flows through a highly modified coastal catchment before entering that is sensitive to sediment and nutrient accumulation. Field observations from the Manly Lagoon excursion indicated a simplified channel form, limited riparian shading in some reaches and visible evidence of urban pressures around the creek corridor. The report assesses the potential effects of two rehabilitation alternatives, which change the channel shape and roughness, on the bankfull velocity, discharge and related suspended sediment and nutrient loads in the study reach. It also determines which alternative is best and other restoration measures that can enhance geomorphology, which slows down the flow, retains sediments and nutrients and promotes a high quality of water downstream.

Sites and methods

This study focusses on Brookvale Creek, a tributary of Manly Lagoon (Figure 1), which has historically had poor water quality, poor geomorphic condition, and poor ecological health. A desktop assessment of two rehabilitation options was conducted to determine how changing channel shape and vegetation roughness in a section of the stream is likely to affect the hydrology, suspended sediment and nutrient loads in the system. Observations, notes and photographs from Manly Lagoon are used to support the data analysis.

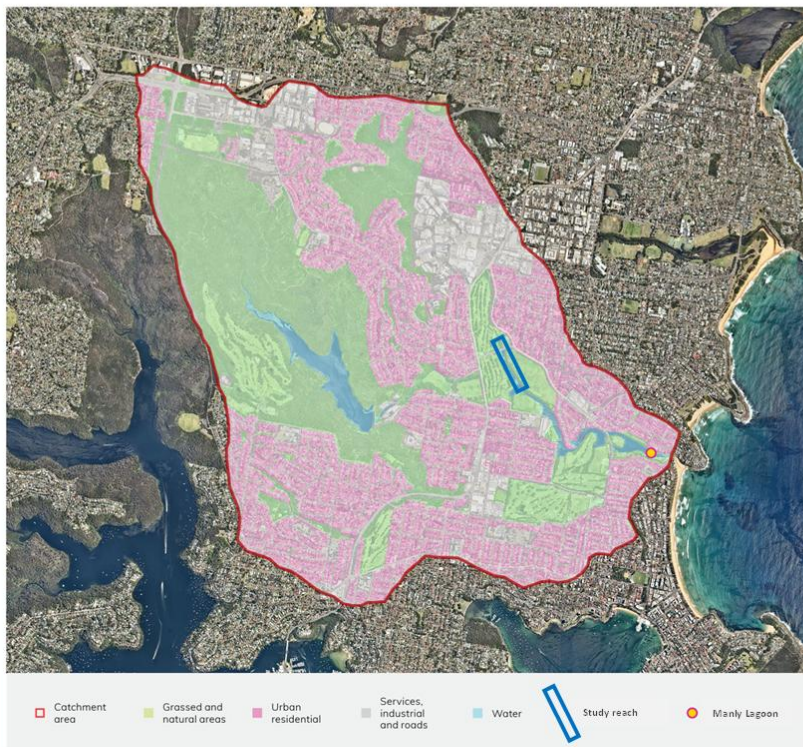


Figure 1. Manly Lagoon catchment, including areas of grassed and natural areas (parks, golf courses, etc.), urban residential areas, industrial areas and roads, waterbodies, and the study reach for proposed rehabilitation options upstream of Manly Lagoon (modified from: Northern Beaches Council 2023, Lagoon Health Report Card 2011-22).

The equations in Table 1 and the information in Table 2 were used to calculate bankfull flow velocity and discharge, suspended sediment and nutrient load estimates, and potential changes from baseline for the two rehabilitation options, A and B:

Option A = a rehabilitation condition with a wider channel and 5-times more roughness than baseline.

Option B = a rehabilitation condition with a wider and shallower channel and 10-times more roughness than baseline.

Table 1. Equations used for hydrology and load estimates.

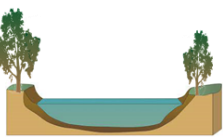
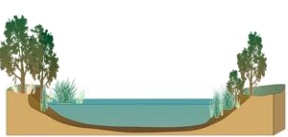
Equation 1: Flow velocity	Equation 2: Discharge	Equation 3: Load
$V = [(D^{2/3}) \times (S^{1/2}) \div n]$	$Q = V \times A$	$L = Q \times C$
Where: V = mean channel velocity (m/s) D = flow depth (m) S = water surface slope (m/m; approx. equal to bed slope) n = Manning's roughness coefficient	Where: Q = discharge (m ³ /s) V = velocity (m/s) A = cross-sectional area (m ²)	Where: L = load (mg/s) Q = discharge (m ³ /s) C = concentration (µg/L) <i>Note: To convert load (mg/s) to load (t/yr), divide the value by 31.71</i>

Results

Table 2 demonstrates that a significant decrease in bankfull velocity and discharge was caused by an increased channel roughness compared with the existing state. The calculated velocity (2.75 m/s) and discharge (22.00 m³/s) were the highest in the baseline channel. Option A that increased the channel and multiplied n of Manning fivefold decreased velocity to 0.55 m/s and discharge to 6.60 m³/s. This is an 80.0% velocity decrease and a 70.0% discharge decrease compared to baseline. B gave the highest change: the velocity decreased to 0.24 m/s and the discharge decreased to 3.84 m³/s, which is 91.4 and 82.8 percent, respectively.

Since the upstream concentrations were assumed to be constant, the load computations reflected the discharge pattern. In the baseline conditions, the annual loads were estimated at 4856.25 t/yr total suspended solids, 174.18 t/yr phosphate and 2970.21 t/yr nitrate. These fell to 1456.88, 52.25 and 891.06 t/yr under Option A and 837.00, 30.02 and 511.93 t/yr under Option B. The two rehabilitation designs are therefore projected to decrease downstream sediment and nutrient transportation with the most significant decrease at 837.00. As noted in Figure 1, the two rehabilitation alternatives reduce hydraulic power drastically compared with the baseline, and the decrease in constituent loads (Figure 2) is in the same order as the decrease in discharge.

Table 2. Schematic channel cross-sections, long-term water quality conditions, channel metrics, and estimates of flow velocity, discharge, suspended sediment and nutrient loads.

Baseline and Options	Current condition (baseline, narrow with minimal roughness)	Option A – rehab. condition (wider with 5x more roughness)	Option B – rehab. condition (wider and shallower with 10x more roughness)
Schematic of channel cross-section			
Current water quality conditions (long-term average, assume no change for options A and B)[†]			
Temperature (°C)	17.3	17.3	17.3
pH	6.9	6.9	6.9
Electrical conductivity (EC; µS/cm)	1050	1050	1050
Dissolved oxygen (DO; %)	53.2	53.2	53.2
Turbidity (NTU)	14.6	14.6	14.6
Total suspended solids (TSS; µg/L)	7001	7001	7001
Total phosphorous (TP; µg/L)	81.8	81.8	81.8
* Phosphate (PO ₄ ; µg/L)	251.1	251.1	251.1
Total nitrogen (TN; µg/L)	966.6	966.6	966.6
[^] Nitrate (NO ₃ ; µg/L)	4282	4282	4282
Channel morphology and hydrology			
Bankfull width (m)	4	6	10
Bankfull depth (m)	2	2	1.6
Bankfull cross-sectional area (m ²)	8	12	16
River bed slope (m/m)	0.0003	0.0003	0.0003
Manning's n roughness coefficient	0.01	0.05	0.1
Bankfull velocity (m/s)	2.75	0.55	0.24
Bankfull discharge (m ³ /s)	22.00	6.60	3.84
Suspended sediment and nutrient loads			
# Total suspended solid load (t/yr)	4856.25	1456.88	837.00
# Phosphate load (t/yr)	174.18	52.25	30.02
# Nitrate load (t/yr)	2970.21	891.06	511.93
Potential changes in hydrology and loads from baseline			
Change in velocity from baseline (%)	n/a	-80.0	-91.4
Change in discharge from baseline (%)	n/a	-70.0	-82.8
Change in total suspended solid load from baseline (%)	n/a	Same as % change in discharge	Same as % change in discharge
Change in phosphate load from baseline (%)	n/a	Same as % change in discharge	Same as % change in discharge
Change in nitrate load from baseline (%)	n/a	Same as % change in discharge	Same as % change in discharge

Notes:

[†] Data adapted from: Warringah and Manly Councils with Patterson Britton & Partners (2003) Manly Lagoon rehabilitation works sediment removal from Sites 3 and 5, Statement of Environmental Effects, Issue 2, Aug 2003.

* To convert total phosphorous (TP; µg/L) to phosphate (PO₄; µg/L), multiply the value by 3.07.

[^] To convert total nitrogen (TN; µg/L) to nitrate (NO₃; µg/L), multiply the value by 4.43.

To convert load (mg/s) to load (t/yr), divide the value by 31.71.

Additional figures or tables:

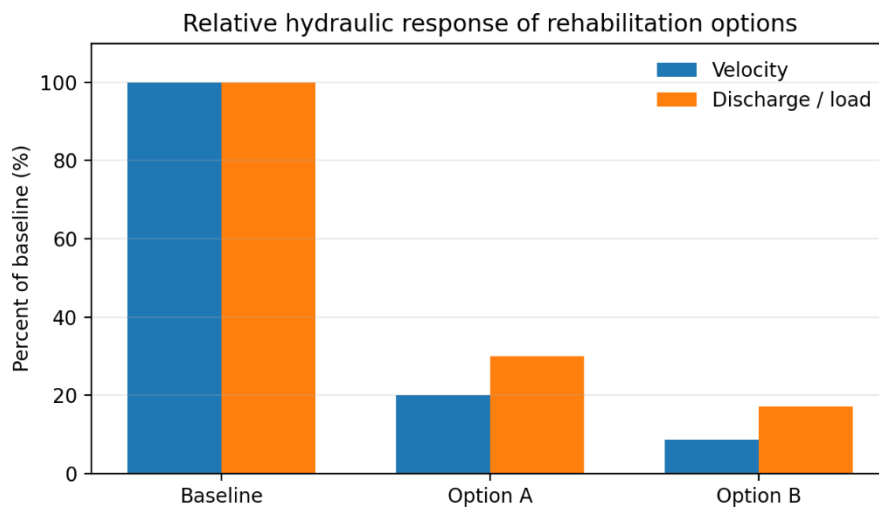


Figure 1. *Relative change in bankfull velocity and discharge*

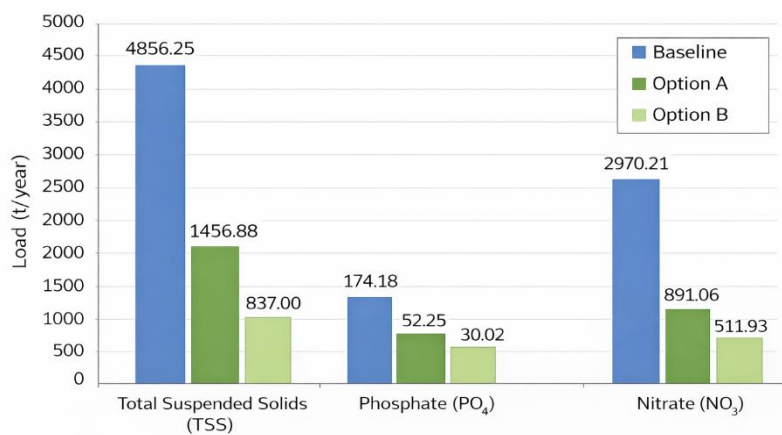


Figure 2. *Comparison of TSS, phosphate (PO_4), and nitrate (NO_3) loads under baseline and rehabilitation scenarios*



Figure 3: *Turbid water discharge from Manly Lagoon into the ocean*

Discussion

Option B is more robust rehabilitation alternative since it provides the most significant changes in both bankfull velocity and discharge, which are likely to decrease the erosive force, enhance sediment retention and reduce the downstream conveyance of suspended solids and dissolved nutrients. In urban streams, incision, bank erosion, habitat simplification and water-quality degradation are largely promoted by storm flows which are often rapid and frequent (Paul and Meyer, 2001; Walsh et al., 2005; Chin et al., 2013). Option B diffuses flow over a greater area by expanding the channel, making it shallower and much rougher, which causes more hydraulic resistance due to vegetation and woody roughness. This mixture is superior to widening by itself to slow water and douse the flashiness that defines runoff in urban areas (Booth, 2000; Burns et al., 2012).

Option A would still be a significant improvement and could be easier to build or repair, but its cross-section is deeper, resulting in water being more concentrated and faster than in Option B. That probably results in increased residual capacity to bed and bank erosion, sediment entrainment and downstream nutrient export. However, channel redesign by itself is rarely enough to restore urban streams if the catchment continues to deliver flashy runoff and pollutants from impervious surfaces (Bernhardt & Palmer, 2007; Walsh et al., 2012). For this reason, the preferred strategy is to implement Option B together with catchment-scale stormwater controls.

Other rehabilitation measures must encompass riparian revegetation using native plants, reconnection of small benches in the floodplains where space is available, gross pollutant traps and sediment forebays installed upstream, biofiltration systems like raingardens and vegetated swales and improved erosion and nutrient control in parks, roadside and golf-course drainage banks. Covering or augmenting forested and vegetated cover in the catchment would also aid in restoring infiltration and decreasing volumes of runoff at the source (Booth, 2000). Increasing urbanisation has been demonstrated to severely impair the quality of water in streams and lagoons whereas coastal lagoons need management specific to their catchment because they store the impact of disturbance upstream over time (Preston, 1995; Ruszczyk et al., 2014; Birch et al., 2021). Thus, the most justifiable suggestion is that of Option B with an incorporated stormwater and riparian restoration throughout the broader catchment.

Conclusion

The evaluation indicates that flow energy and pollutant transport in Brookvale Creek can be significantly decreased by increasing the channel roughness and re-forming. The most successful design is option B, which will only be able to offer long-lasting improvement to Manly Lagoon through in-channel rehabilitation of floodplains with stormwater, riparian and sediment control tools on a catchment scale.

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